Discussion of

How Does Relaxing Funding Constraints Affect Markets? Evidence from Flash Loans

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What the paper does

- **Question:** what happens to markets when funding constraints are relaxed?
- **Setting:** flash loans — uncollateralised loans that must be repaid within the same blockchain transaction
  - Aave's launch (8 Jan 2020) made 10 utility tokens flash-loanable
  - Compare 167 other tokens' pools
- **Part 1 — causal:** pool depth rises ≈ 0.72 log points; no deterioration in price impact or volatility; new liquidity supplied by entrant LPs
- **Part 2 — anatomy:** all 212,690 flash loans, 2020–2024
  - Uses: debt position management (48%), arbitrage (35%), attacks (<0.3%)
  - Arbitrage: flash funding used for trades large relative to the operator's wealth
  - Borrowers: use it near the liquidation threshold; voluntary exits leave a smaller price footprint than forced liquidations

Overview

What I liked about the paper

Questions about the experiment

1. Are the treated and control pools comparable?
2. Does the timing line up with the treatment?
3. Early 2020 was not a normal period

Question about external validity

Presentation and accessibility

Looking ahead: two papers?

What I like

- **An important, new funding technology**, and a creative attempt to study it
- **Careful implementation with several methodologies**: wild cluster bootstrap, randomization inference, synthetic DiD (MacKinnon and Webb (2018), Arkhangelsky et al. (2021))
- **Clever use of AMM mechanics**: depth $= \sqrt{Kp}$ splits the effect into new liquidity vs. price appreciation
- **A massive data effort**: every flash loan, every lending position, complete transaction data for six months
- **A useful map of what flash loans are used for**: mostly debt management and arbitrage; confirmed attacks are rare

1. Are the treated and control pools comparable?

Variable	Treated mean	Treated median	Control mean	Control median	Norm. diff.
Token-day					
ETH depth (ETH reserve)	1,990.3	636.4	51.7	1.9	0.85
Volume (ETH per day)	735.0	52.9	4.6	0.0	0.47
Swaps per day	100.7	42.0	3.1	0.0	0.85
Mean trade size (ETH), trading days	2.7	1.3	1.5	0.3	0.16
Unique traders per day	27.0	15.5	1.6	0.0	1.20
Mean price impact per trade (bp)	64.4	22.9	316.8	97.4	-0.26
Volatility of swap-to-swap price changes (bp)	116.7	58.9	312.5	128.8	-0.39

Table 4.1 of the paper (token-day panel, pre-period).

- Median depth 636 vs. 2 ETH; the median control pool **does not trade** on a typical day.
- Fixed effects remove differences in *levels*, not in *dynamics*: large and tiny pools need not grow alike
- Author addresses with hand-picked comparable controls, but we do not see the covariate balance for that subset

Suggestion: match or re-weight controls on size and activity; consider the comparable-controls estimate as the headline (with balance table).

2. Does the timing line up with the treatment?

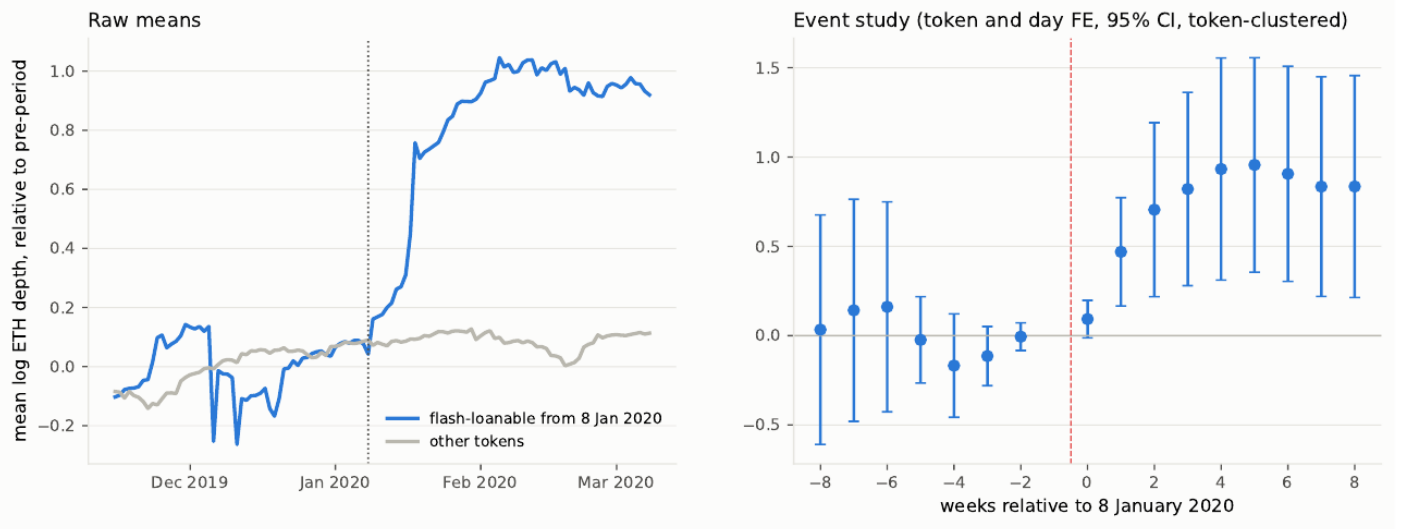


Figure 4.1 of the paper. Launch: 8 January 2020 (dotted line).

- Treated pools move well *before* the launch: a rise into early December, then sharp one-day drops
- The main rise is a near-vertical step roughly two weeks *after* the launch

Very few flash loans made in January

Month	Market state	ETH return	Largest drawdown	Transactions	Arbitrages (flash-funded)	Flash loans	Liquidations	Self-exits (flash-funded)
2019-12	before Aave launch	-15%	-19%	18,271,815	8,104 (0)	0	321	46 (0)
2020-01	Aave launch	+38%	-8%	17,703,536	9,767 (0)	7	252	104 (1)
2020-03	crash (12 March)	-39%	-55%	22,748,700	19,442 (0)	1,490	1,888	995 (450)
2021-05	crash (19 May)	-8%	-50%	45,055,042	215,670 (185)	9,214	6,788	7,231 (3,611)
2022-06	crash (Celsius, 3AC)	-41%	-46%	31,083,501	124,987 (620)	5,801	9,755	3,044 (2,040)
2023-09	quiet	+3%	-6%	30,490,755	85,795 (1,806)	7,370	550	896 (542)

2. Timing (continued): what moved the treated pools?

If almost no one used flash loans in January, what is the channel?

- The paper only suggests answer: LPs anticipate new order flow
- But that needs evidence, preferably more closely related to the timing of the intervention

Suggestion: More granular view of flashloan uptake and tighter linkage to intervention timing / mechanism.

3. Early 2020 was not a normal period

- The window (15 Nov 2019 – 8 Mar 2020) spans an extraordinary stretch
  - Jan – mid-Feb 2020: sharp crypto rally; ETH roughly doubled; large DeFi tokens rallied strongly
  - Late Feb: the COVID turn begins, ahead of the 12 March crash

- Day fixed effects absorb *common* shocks only if treated and control pools load on them equally
  - Treated: large, liquid, DeFi-flavoured tokens
  - Control: mostly tiny pools that rarely trade
  - Differential exposure to the rally and to DeFi attention looks like a treatment effect

Suggestion: control for token-level beta and momentum; run a placebo on large DeFi tokens that are not flash-eligible.

4. External validity: a very specific setting

Title: *How Does Relaxing **Funding Constraints** Affect **Markets**? Evidence from Flash Loans*

- The introduction motivates with 2008 margin spirals, the COVID "dash for cash," and Treasury stress
- Limits-to-arbitrage models (Shleifer and Vishny (1997), Gromb and Vayanos (2002)) concern capital tied up *over time*, which a flash loan cannot replicate

Suggestion: match the title and framing to the evidence (e.g., "Flash Loans and Liquidity in Decentralised Markets"); present the broader implications as hypotheses.

5. Presentation and accessibility

- 35 exhibits make the main contribution hard to see (more on this at the end)
- Help a finance audience into the setting: define health factors, atomicity, etc. early
- Tidy internal consistency
  - "Causal" language in places vs. "we do not interpret causally" in others
  - "We" for a solo authored paper

Looking ahead: two papers

An interesting question, a novel setting, and an impressive data effort — but it reads as two papers

1. The experiment

January, 2020 Launch of Flash Loans

- Show genuinely comparable controls
- Line the timing up with actual flash-loan use
- Rule out differential exposure to other early-2020 shocks
- Narrow the claims to the setting

2. The anatomy

Who uses instantaneous funding, for what, and who bears the consequences?

- Map of uses; arbitrage entry? forced vs. voluntary sales? Attacks vs benign use?

- If the experiment survives scrutiny, both papers stand; if not, the anatomy is a promising paper on its own
- I look forward to the next version(s)